RAHATLINK

SMART DISASTER RELIEF INVENTORY OPTIMIZER

Connecting Relief, Saving Lives

AI-Powered | Real-Time | Verified Data | Optimized

Hackathon Project — AI FOR A BETTER WORLD — Disaster Relief Tech

CATEGORY	TECH STACK	TEAM
Disaster Management & AI	React · FastAPI · PostgreSQL · Leaflet.js	Rahatlink Team

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■ 01 — INTRODUCTION

What is RahatLink?

RahatLink is an intelligent disaster relief platform that connects warehouses and NGOs through real-time location data, verified disaster information, and smart allocation logic. It ensures efficient, transparent, and prioritized distribution of essential supplies during emergencies.

■ 02 — PROBLEM STATEMENT

The Core Challenges in Disaster Relief

Current disaster relief operations are plagued by systemic inefficiencies that cost lives and waste resources.

#	Problem	Impact
1	Lack of coordination between stakeholders	Delayed aid delivery
2	Fake or unverified disaster reports	Resource misallocation
3	Delayed response times	Increased casualties
4	Unequal resource distribution	Underserved critical zones
5	No real-time supply-demand visibility	Operational blindness

■ 03 — SOLUTION OVERVIEW

How RahatLink Solves the Problem

RahatLink provides a unified platform that addresses every gap in the current disaster relief ecosystem through five core pillars:

- **Verified Disaster Identification**
  A dedicated News Page pulls confirmed disaster updates, ensuring only legitimate zones receive priority status.
- **Priority-Based Allocation**
  Disaster zones receive highest priority. Resources flow to the most critical areas first using urgency-based sorting.
- **Live Map Visualization**
  All warehouses and camps are displayed in real-time on an interactive map with clickable markers and route lines.
- **Distance-Based Intelligent Routing**
  The system identifies the nearest warehouse to each disaster camp and calculates optimal supply routes.
- **Manual + Automated Distribution**
  Coordinators can manually select or trigger the automated optimization engine for system-wide allocation.

■ 04 — DISASTER VERIFICATION SYSTEM

The News Page is the cornerstone of RahatLink's integrity. It acts as a real-time gate that only allows confirmed disaster zones to enter the priority pipeline, preventing misuse of limited resources.

Feature	Description
Dedicated News Page	Displays curated, verified disaster updates from reliable sources
Zone Verification	Only confirmed disaster locations are marked as high-priority zones
Misuse Prevention	Unverified requests are blocked from accessing priority resources
Priority Engine Driver	Verified zones directly feed into the allocation priority system

■ 05 — PRIORITY SYSTEM

Three-Tier Priority Architecture

TIER	ZONE TYPE	CRITERIA	PRIORITY LEVEL
1	Active Disaster Zones	Verified by News Page	■ CRITICAL
2	High-Urgency Camps	Urgency flag set by NGO	■ HIGH
3	Standard Relief Areas	Non-disaster, normal ops	■ NORMAL

■ 06 — COMPLETE WEBSITE STRUCTURE

The platform is composed of 11 interconnected pages, each serving a distinct role:

#	Page Name	Primary Role	Users
01	Landing Page	Product introduction & entry point	All
02	Role Selection Page	Choose Warehouse or NGO role	All
03	Location Setup Page	GPS or manual location entry	All
04	Main Dashboard	Statistics, charts & system overview	All
05	Map View Page	Interactive entity visualization	All
06	Warehouse Panel	Manage & update inventory	Warehouses
07	NGO Panel	Post and manage needs/requests	NGOs
08	Distribution Page	Manual supply allocation	Coordinators
09	Optimization Page	AI-driven auto-allocation engine	Coordinators
10	News Page	Disaster verification hub	All
11	Analytics Dashboard	Charts, tables & allocation data	All

■ 07 — PAGE-BY-PAGE UI/UX DESIGN

PAGE 01 — Landing Page

UI Elements	Interactions
■ Dark modern hero section with title 'RahatLink'	■ Single click CTA navigates to role selection
■ Tagline: 'Connecting Relief, Saving Lives'	■ Smooth scroll animations between sections
■ CTA button: 'Get Started' (prominent, orange)	■ Hover effects on feature cards
■ Features overview section with icon cards	
■ System benefits and demo preview cards	

PAGE 02 — Role Selection Page

UI Elements	Interactions
■ Two large centered cards: Warehouse & NGO	■ Click to select role → data stored in session
■ Icon + label for each role type	■ Smooth slide transition to Location Setup
■ Hover glow effect on cards	
■ Clean minimal layout	

PAGE 03 — Location Setup Page

UI Elements	Interactions
■ Full-screen centered map view	■ Auto GPS detection places marker on map
■ 'Detect My Location' GPS button	■ Location coordinates stored for future routing
■ Manual address input field below map	■ Manual input geocodes address to coordinates
■ Animated location pin marker	

PAGE 04 — Main Dashboard

UI Elements	Interactions
■ Collapsible sidebar navigation	■ Stat cards update on every allocation
■ Top stat cards: Total Supply · Total Demand · Alerts	■ Charts re-render with live data

Supply vs Demand Bar Chart	Sidebar links navigate between pages
Resource Distribution Pie Chart	

PAGE 05 — Map View Page (CORE)

UI Elements	Interactions
Full-screen Leaflet.js interactive map	Click warehouse marker → popup with inventory & distance
Warehouse markers (blue pins)	Click camp marker → popup with needs & urgency
Camp / NGO markers (red pins)	Select warehouse + camp → draw route line
Disaster zone highlight overlay	Filter buttons: show only warehouses / camps / all
Route line between selected entities	

PAGE 06 — Warehouse Panel

UI Elements	Interactions
Add Stock form: Item type + Quantity	Add / Update stock → database updated
Item types: Food, Water, Medical Kits, Essentials	Inventory reflects across Dashboard & Map
Current inventory table with Edit/Delete	Form validation on empty/invalid input
Total stock summary card	

PAGE 07 — NGO Panel

UI Elements	Interactions
Post Need form: Item + Quantity + Urgency level	Submit need → enters allocation queue
Urgency: Low / Medium / High / Critical	Urgency level influences priority scoring
Current needs table	Status auto-updates after distribution
Fulfillment status badges (Pending/Partial/Fulfilled)	

PAGE 08 — Distribution Page

UI Elements	Interactions

Dropdown: Select Warehouse	Submit → deducts from warehouse stock
Dropdown: Select Camp	Camp need quantity reduced accordingly
Dropdown: Select Item	Dashboard & Analytics updated in real-time
Number input: Quantity	
'Send Supplies' action button	

PAGE 09 — Optimization Page

UI Elements	Interactions
'Optimize Distribution' trigger button	Runs greedy distance + priority algorithm
Algorithm run indicator / loading state	Displays complete allocation plan
Result table: Warehouse → Camp → Item → Qty	One-click to confirm and apply all allocations
Accept / Modify allocation options	

PAGE 10 — News Page (CORE)

UI Elements	Interactions
Grid of disaster news cards	Verified disaster cards auto-mark locations as priority
Each card: Title · Location · Date · Status badge	Click card to see full disaster details
Verified badge in green, Unverified in grey	Data synced to map overlay and priority engine
Filter: All / Verified / By Region	

PAGE 11 — Analytics Dashboard

UI Elements	Interactions
Bar chart: Supply vs Demand by category	Charts auto-refresh on data changes
Pie chart: Resource distribution across camps	Date range filter for historical analysis
Allocations history table	Export to CSV option for coordinators
Remaining stock table · Unfulfilled needs table	

■ 08 — INTERACTIVITY & USER FLOW

Key Interactive Features

Feature	Description	Trigger
Real-Time Map Interaction	Clickable markers with live popup data	Map click
Route Visualization	Animated line between warehouse and camp	Entity selection
Dynamic Graph Updates	Charts re-render after every allocation	Data change
Priority Badge System	Red/Orange/Green badges on map markers	Zone verification
Optimization Engine	One-click full-system allocation run	Button click
News Verification Feed	Disaster cards link directly to priority map	Card verification
Role-Based Views	Dashboard adapts to Warehouse or NGO role	Login/role select

■ 09 — SYSTEM WORKFLOW

End-to-End Operation Flow

01	**Role Selection** User visits RahatLink and chooses between Warehouse or NGO role.
02	**Location Detection** GPS auto-detect or manual address input stores user coordinates.
03	**News Page Verification** Coordinators review disaster news; verified zones marked as high priority.
04	**Warehouse Inventory Entry** Warehouse managers add available stock: Food, Water, Medical Kits, Essentials.
05	**NGO Needs Posting** Field workers submit needs with item type, quantity, and urgency level.
06	**Map Entity Visualization** All warehouses and camps appear as interactive map markers.
07	**Priority Zone Application** System flags disaster zones; urgency sorting applied within zones.
08	**Manual or Automated Distribution** Coordinators manually allocate or run the optimization engine.
09	**Allocation Generation** System generates warehouse → camp → item → quantity allocation plan.
10	**Dashboard Real-Time Update** All statistics, charts, and tables update instantly post-allocation.

■■ 10 — DATABASE DESIGN

Relational Database Schema (PostgreSQL / Supabase)

Table	Key Fields	Relationships
users	id, name, email, role, location_lat, location_lng	Links to warehouses, camps
warehouses	id, user_id, name, lat, lng, address	Has many inventory items
inventory	id, warehouse_id, item_type, quantity, updated_at	Belongs to warehouse
camps	id, user_id, name, lat, lng, is_disaster_zone, urgency	Has many needs
needs	id, camp_id, item_type, quantity, urgency, status	Belongs to camp
allocations	id, warehouse_id, camp_id, item_type, qty, created_at	References both entities
news	id, title, location, lat, lng, is_verified, date	Links verified zones to camps

■ 11 — API DESIGN

FastAPI RESTful Endpoints

Method	Endpoint	Purpose
POST / GET	/api/auth/register, /login	User registration and authentication
GET / POST	/api/warehouses	List or create warehouses
GET / PUT	/api/inventory/{warehouse_id}	Fetch or update warehouse stock
GET / POST	/api/camps	List or register NGO camps
GET / POST	/api/needs/{camp_id}	Fetch or submit camp needs
POST / GET	/api/allocations	Create or view distributions
POST	/api/optimize	Run optimization algorithm
GET / POST	/api/news	Fetch or submit disaster news
GET	/api/map/entities	All map markers with location data
GET	/api/analytics/summary	Dashboard stats and chart data

■ 12 — ALGORITHM LOGIC

Smart Allocation Algorithm

The optimization engine uses a priority-weighted greedy algorithm that sorts camps by disaster status and urgency, then matches each camp with the nearest warehouse that has sufficient stock.

Step 1	**Load & Classify**	Load all camps. Flag disaster zones (verified via News Page). Mark urgency levels.
Step 2	**Priority Sort**	Sort camps: (1) Disaster zones first, (2) Urgency level descending, (3) Alphabetical tie-break.
Step 3	**Nearest Warehouse**	For each camp, calculate Haversine distance to all warehouses with available stock.
Step 4	**Greedy Allocation**	Allocate from the nearest warehouse. Deduct from warehouse inventory. Mark need as fulfilled or partial.
Step 5	**State Update**	After all allocations, update database. Return allocation plan for display.

■■ 13 — TECHNOLOGY STACK

Frontend

React (Vite)	Fast modern frontend build tool
Tailwind CSS	Utility-first styling framework
ShadCN UI	Accessible component library
Chart.js	Bar and pie data visualizations
Leaflet.js	Interactive maps with custom markers

Backend

FastAPI (Python)	High-performance async REST API
Pydantic	Data validation and schemas
SQLAlchemy	ORM for database interaction
Haversine	Distance calculation library

Database & Infra

PostgreSQL	Relational database engine
Supabase	Managed Postgres + Auth + Storage
PostGIS	Geospatial data extensions

■ 14 — UI/UX DESIGN PRINCIPLES

Design Philosophy

- **Dark Theme**
  Deep #0D1117 background with card panels at #161B22 reduces eye strain during prolonged emergency operations.
- **Clean Spacing**
  Generous padding and white space ensure readability under stress. No cluttered elements.
- **Card-Based Layout**
  Information organized in discrete cards makes scanning quick. Each card has a single clear purpose.
- **Visual Hierarchy**
  Font sizes, colors, and weight guide the eye from most to least critical information.
- **Interactive Feedback**
  Every action produces visible feedback: button states, loading spinners, success toasts.
- **Minimal & Intuitive**
  No training required. Field workers and NGO coordinators can operate the system immediately.

■ 15 — CONCLUSION

The RahatLink Vision

RahatLink is a scalable and intelligent platform that fundamentally transforms disaster response operations. By combining verified information pipelines, real-time location intelligence, and an optimized distribution engine, RahatLink ensures that critical resources reach the most vulnerable populations quickly and reliably.

Impact Summary

Impact Area	Without RahatLink	With RahatLink
Verification	Fake reports waste resources	Only verified zones receive priority aid
Coordination	Silos between NGOs & warehouses	Unified real-time platform for all
Routing	Manual, distance-unaware allocation	Nearest warehouse auto-selected
Speed	Hours of manual coordination	Allocation generated in seconds
Visibility	No overview of supply-demand	Live dashboard with full transparency
Fairness	Ad-hoc, inequitable distribution	Priority-based, data-driven allocation

RahatLink is not just a hackathon project — it is a blueprint for saving lives through technology.

Built with compassion. Engineered for impact.